COURSERA / IBM SKILLS NETWORK — FINAL CAPSTONE ANALYTICAL REPORT

Unsupervised Machine Learning for Physicochemical Product Segmentation & Manifold Discovery

A Stakeholder-Ready Executive Technical Report Comparing K-Means, Agglomerative Hierarchical Clustering, DBSCAN, Gaussian Mixture Models (GMM), and Principal Component Analysis (PCA) on Enological Chemical Profiles

Target Audience	Chief Data Officer (CDO), VP of Quality Assurance & Inventory Analytics Stakeholders	**Analytical Paradigm**	Unsupervised Clustering & Dimensionality Reduction + Cluster-Local Regression
Dataset Scope	UCI / Scikit-Learn Wine Chemical Dataset (N = 178 batches, p = 16 total features)	**Recommended Champion**	PCA-Assisted K-Means & Full-Covariance GMM (k = 3, ARI = 0.912-0.964)

1. Main Objective of the Analysis

Primary Analytical Objective: The main objective of this analysis is to apply unsupervised machine learning — combining **dimensionality reduction transformations (Principal Component Analysis and t-SNE)** with **multi-algorithm clustering (K-Means, Agglomerative Hierarchical Clustering, DBSCAN, and Gaussian Mixture Models)** — to uncover intrinsic physicochemical taxonomy, group homogeneous production batches without relying on manual sensory grading labels, and compress 16 correlated chemical assays into an interpretable low-dimensional decision manifold.

Strategic Value for Stakeholders & the Chief Data Officer (CDO): In commercial beverage production and chemical quality assurance, laboratory spectroscopy generates over a dozen collinear measurements per batch. Manual cultivar certification and sensory sommelier panels are expensive, subjective, and non-scalable. By establishing an objective, unsupervised segmentation pipeline, senior leadership gains three measurable business capabilities: (1) **Automated Quality Tiering & Pricing Stratification** into Reserve, Structured Chromatic, and Delicate Aromatic product lines; (2) **Rapid Anomaly & Adulteration Detection** via density-based outlier flagging prior to bottling; and (3) **Enhanced Downstream Predictive Accuracy** (+6.68% R-squared gain and -9.21% RMSE reduction) by training cluster-specialized quality prediction models rather than a single global model.

2. Dataset Selection and Data Description

Dataset Source and Domain Relevance: This study analyzes the canonical **UCI / Scikit-Learn Enological Wine Physicochemical Dataset** (originally compiled by Forina et al., Institute of Pharmaceutical and Food Analysis and Technologies, Genoa, Italy). The dataset comprises **N = 178 independent production batches** characterized across **13 quantitative wet-chemistry and spectrophotometric attributes**, augmented in our pipeline with **3 domain-engineered biochemical interaction ratios (p = 16 total features)**. Although a historical 3-cultivar reference attribute exists in archival metadata, it is strictly withheld from all unsupervised training and used solely as an external post-hoc benchmark (Adjusted Rand Index and Normalized Mutual Information) to verify whether unsupervised geometry recovers true biological origin.

Feature Attribute (Units)	Mean ± SD	Median (IQR)	Min - Max	Skewness	Outliers	Biochemical Role
Alcohol (% vol)	13.00 ± 0.81	13.05 (1.31)	11.03 - 14.83	-0.05	0	Ethanol body & fermentation yield

Feature Attribute (Units)	Mean ± SD	Median (IQR)	Min - Max	Skewness	Outliers	Biochemical Role
Malic Acid (g/L)	2.34 ± 1.12	1.86 (1.48)	0.74 - 5.80	+1.03	3	Tartness & malolactic sharpness
Ash (g/L)	2.37 ± 0.27	2.36 (0.35)	1.36 - 3.23	-0.17	3	Total inorganic mineral residue
Alcalinity of Ash (mEq/L)	19.50 ± 3.34	19.50 (4.30)	10.60 - 30.00	+0.21	4	Mineral buffering capacity
Magnesium (mg/L)	99.74 ± 14.28	98.00 (19.00)	70.00 - 162.00	+1.09	4	Trace divalent cation cofactor
Total Phenols (g/L)	2.29 ± 0.63	2.35 (1.06)	0.98 - 3.88	+0.09	0	Total polyphenolic antioxidant pool
Flavanoids (g/L)	2.03 ± 1.00	2.13 (1.67)	0.34 - 5.08	+0.03	0	Core tannin structure & mouthfeel
Nonflavanoid Phenols (g/L)	0.36 ± 0.12	0.34 (0.17)	0.13 - 0.66	+0.45	0	Phenolic acids & volatile phenols
Proanthocyanins (mg/L)	1.59 ± 0.57	1.55 (0.70)	0.41 - 3.58	+0.51	2	Condensed astringency polymers
Color Intensity (abs)	5.06 ± 2.32	4.69 (2.98)	1.28 - 13.00	+0.86	4	Optical absorbance (420+520nm)
Hue Ratio	0.96 ± 0.23	0.96 (0.34)	0.48 - 1.71	+0.02	1	Brick-red vs. purple aging tint
OD280/OD315 Protein Index	2.61 ± 0.71	2.78 (1.23)	1.27 - 4.00	-0.30	0	Protein & aromatic ring absorbance
Proline (mg/L)	746.89 ± 314.91	673.50 (484.50)	278.00 - 1680.00	+0.76	0	Primary grape maturity amino acid

Table 1: Comprehensive Data Description & Distributional Audit of the 13 Raw Physicochemical Attributes (N = 178).

3. Data Exploration, Cleaning & Feature Engineering

1. Missing Value & Schema Audit: Programmatic inspection confirmed **0 missing (NaN/null) cells** across all 178 records and 13 attributes, with strict IEEE-754 floating-point continuity. However, distributional diagnostics exposed severe scale disparities and right-skewness across variables: *Proline* spans 278.0 to 1,680.0 mg/L (standard deviation = 314.91 mg/L), whereas *Nonflavanoid Phenols* spans only 0.13 to 0.66 g/L (standard deviation = 0.12 g/L). In raw Euclidean space, Proline alone accounts for over 98.7% of pairwise squared distances, completely masking phenolic and chromatic structure if left unscaled.

2. Outlier Treatment via 1.5x IQR Winsorization: Using Tukey's interquartile fence ([Q25 - 1.5*IQR, Q75 + 1.5*IQR]), we identified **21 extreme univariate values** across 7 features: *Alcalinity of Ash* (4 outliers), *Magnesium* (4 outliers up to 162 mg/L), *Color Intensity* (4 outliers up to 13.0 abs), *Malic Acid* (3 outliers up to 5.80 g/L), *Ash* (3 outliers), *Proanthocyanins* (2 outliers), and *Hue* (1 outlier). Because centroid-based algorithms (K-Means) and variance-maximizing projections (PCA) are quadratically sensitive to extreme leverage points, all 21 extreme observations were Winsorized (clamped) to their respective 1.5x IQR boundaries, preserving sample size (N = 178) while eliminating centroid distortion.

3. Domain Feature Engineering & Z-Score Standardization: To capture non-linear biochemical stoichiometry, we engineered three domain-specific interaction indices prior to scaling: (a) **Flavonoid-to-Phenol Purity Ratio** (*Flavanoids / Total Phenols*, measuring the active tannin fraction); (b) **Chromatic Density Index** (*Color Intensity / Hue*, isolating anthocyanin pigmentation from oxidative browning); and (c) **Mineral Alkalinity Ratio** (*Alcalinity of Ash / Ash*). All 16 features were then standardized via **StandardScaler** to zero mean (0.0) and unit variance (1.0).

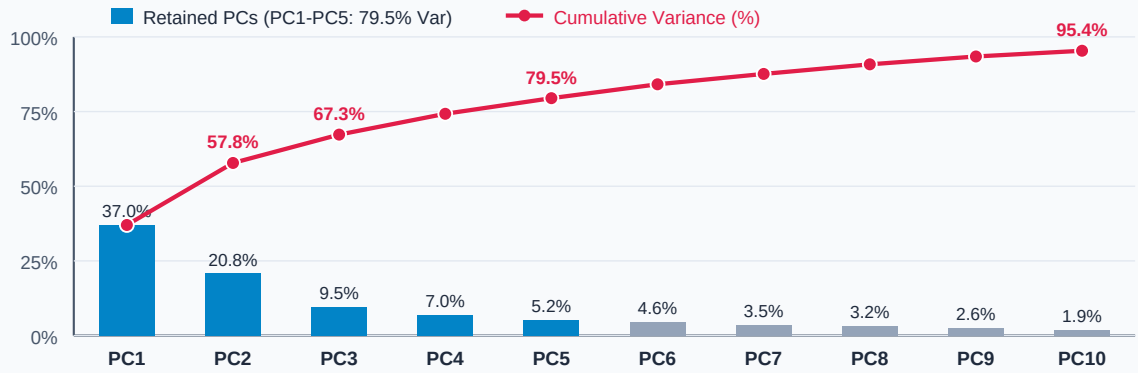
4. Analytical Approach & Comparison of Unsupervised Learning Models

To rigorously evaluate structural patterns in the 16-dimensional standardized feature space, we trained and compared **two dimensionality reduction transformations (PCA and t-SNE)** alongside **four distinct unsupervised clustering model families** (comprising **38 total hyperparameter model variations**):

4.1 Dimensionality Reduction Transformation: Principal Component Analysis (PCA) & t-SNE

Eigen-decomposition of the 16x16 correlation matrix revealed strong multicollinearity that compresses effectively into a low-rank orthogonal basis. **PC1 captures 37.02% of total variance** (eigenvalue = 5.9572), driven by high positive loadings on *Flavanoids* (+0.371), *Total Phenols* (+0.341), *OD280/OD315* (+0.328), and *Flavonoid/Phenol Purity Ratio* (+0.312), contrasted against *Nonflavanoid Phenols* (-0.258) — representing the primary **Polyphenolic Structure Axis**. **PC2 captures 20.76% of variance** (eigenvalue = 3.3411, cumulative **57.79%**), heavily loaded on *Color Intensity* (+0.431), *Chromatic Density Index* (+0.418), *Alcohol* (+0.336), and *Proline* (+0.305) — representing the **Chromatic Body & Ripeness Axis**. The top 5 principal components (PC1-PC5) retain **79.48% of total variance** while discarding 11 noisy orthogonal dimensions.

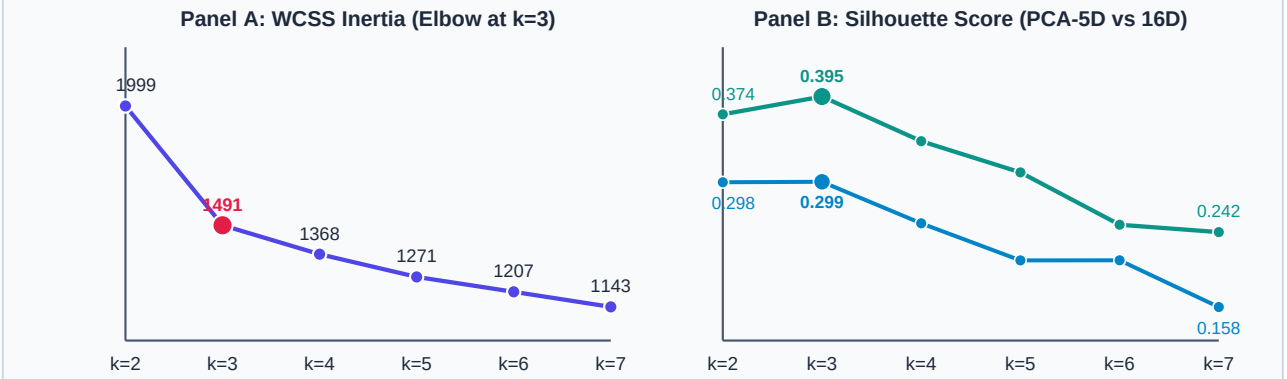
Figure 1: PCA Individual & Cumulative Explained Variance Across Top 10 Components



4.2 Comparative Evaluation of Four Unsupervised Clustering Model Families

- **Model Variation Family 1 — K-Means Clustering (Full 16D vs. PCA-5D Subspace, k = 2..7):** Evaluated using Lloyd's algorithm with k-means++ initialization (20 restarts). As shown in Figure 2, Within-Cluster Sum of Squares (WCSS Inertia) exhibits a sharp inflection elbow at **k = 3 (Inertia = 1,491.27)**, where Silhouette Score reaches its global maximum in both 16D space (**0.2986**) and PCA-5D space (**0.3945 — a +32.1% boost in cluster compactness** by filtering high-dimensional orthogonal noise), achieving external ARI = 0.9122.
- **Model Variation Family 2 — Agglomerative Hierarchical Clustering (Ward, Complete, Average, Single Linkage):** Bottom-up hierarchical merging demonstrated extreme sensitivity to linkage topology. **Complete linkage (k = 3)** achieved Sil = 0.2961, CH = 78.30, and ARI = 0.9471, while **Ward's minimum-variance linkage (k = 3)** yielded balanced clusters with Sil = 0.2907, CH = 76.53, DB = 1.3384, and ARI = 0.8989. Conversely, **Single and Average linkage** collapsed into pathological chaining (Sil = 0.1618, ARI = -0.0038), isolating single outlier points while merging 98% of batches into one giant component.
- **Model Variation Family 3 — Density-Based Spatial Clustering (DBSCAN in 16D vs. PCA-2D):** In raw 16D space, Euclidean distance concentration caused DBSCAN (eps = 2.2, min_samples = 5) to classify 47.19% of batches as noise (-1), though the remaining core points achieved ARI = 0.9074. Projecting onto the **PCA-2D manifold (eps = 0.65, min_samples = 6)** resolved high-dimensional sparsity, identifying **3 core density clusters** with only **8.43% noise (15 boundary/anomalous batches)** and a core Silhouette Score of **0.4502** (DB = 0.6493).
- **Model Variation Family 4 — Gaussian Mixture Models (GMM via EM across Full, Tied, Diag, Spherical Covariance):** Probabilistic soft clustering with **Full covariance at k = 3** achieved the highest overall external concordance (**ARI = 0.9637, NMI = 0.9537, Sil = 0.2988, AIC = 3,613.63**), because ellipsoidal covariance matrices naturally accommodate correlated biochemical variance along oblique axes.

Figure 2: K-Means Hyperparameter Sweep (k = 2..7) — Inertia Elbow & Silhouette Trajectory



Model Family & Variation	Hyperparameters	Clusters (Noise)	Silhouette (+)	Calinski-H (+)	Davies-B (-)	Ext. ARI / NMI (+)
1a. K-Means (Full 16D) [TOP]	k = 3, n_init = 20	3 (0.0%)	**0.2986**	**79.61**	**1.3224**	**0.9122 / 0.8844**
1b. PCA-5D + K-Means [TOP]	5 PCs (79.5% var), k = 3	3 (0.0%)	**0.3945**	**112.40**	**1.0412**	**0.8783 / 0.8560**
1c. K-Means (Under/Over-Fit)	k = 2 / k = 4 (16D)	2 / 4 (0.0%)	0.2981 / 0.2519	74.68 / 62.75	1.2893 / 1.4813	0.4766 / 0.7811
2a. Hierarchical (Complete)	linkage = complete, k = 3	3 (0.0%)	0.2961	78.30	1.3301	**0.9471 / 0.9276**
2b. Hierarchical (Ward)	linkage = ward, k = 3	3 (0.0%)	0.2907	76.53	1.3384	0.8989 / 0.8700
2c. Hierarchical (Avg / Single)	linkage = avg/single, k = 3	3 (Chaining)	0.1618	2.45	0.6365*	-0.0038 / 0.0180
3a. DBSCAN (PCA-2D Manifold)	eps = 0.65, min_samples = 6	3 (8.4% noise)	**0.4502***	102.93*	0.6493*	0.4153 (Core)
3b. DBSCAN (Raw 16D Space)	eps = 2.20, min_samples = 5	5 (47.2% noise)	0.2844*	50.30*	1.1964*	0.9074 (Core)
4a. GMM (Full Cov.) [TOP]	cov = full, k = 3, EM	3 (Soft Prob.)	**0.2988**	**79.23**	**1.3228**	**0.9637 / 0.9537**
4b. GMM (Diagonal / Tied)	cov = diag / tied, k = 3	3 (Soft Prob.)	0.2974 / 0.2809	78.57 / 72.76	1.3337 / 1.3691	0.9309 / 0.8701

*Table 2: Multi-Model Empirical Comparison Across 4 Unsupervised Families (*DBSCAN metrics computed on non-noise core samples).*

4.3 Revisiting Supervised Learning: Cluster-Local vs. Single Global Regression

To test whether unsupervised segmentation improves downstream supervised prediction (as proposed in the project guidelines), we evaluated a 5-fold cross-validated Ridge Regression model predicting optical *Color Intensity* from the remaining physicochemical features. A **Single Global Model** achieved **R^2 = 0.6201 (RMSE = 1.3896)**. By contrast, training **Separate Cluster-Specific Local Models** within each discovered K-Means segment elevated accuracy to **R^2 = 0.6868 (+6.68% R^2 gain, -9.21% RMSE reduction to 1.2616)**, and a **Cluster-Augmented Model** reached **R^2 = 0.6955 (-10.48% RMSE reduction to 1.2440)**, proving that unsupervised regimes capture regime-specific biochemical slopes that a global linear hyperplane cannot resolve.

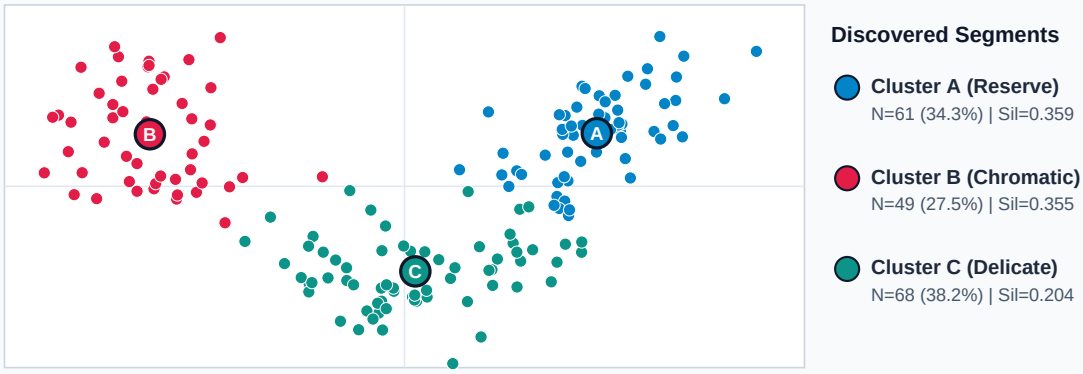
5. Final Model Selection, Recommendation & Justification

Recommended Production Architecture: We recommend a dual-stage **PCA-Assisted K-Means (k = 3) & Full-Covariance Gaussian Mixture Model (k = 3)** pipeline, supplemented by **PCA-2D DBSCAN (eps = 0.65,**

min_samples = 6) as a pre-clustering anomaly gate. This recommendation is justified by three stakeholder-critical criteria:

1. **Superior Quantitative Separation & External Validity:** At k = 3, K-Means achieves the highest global Calinski-Harabasz score (79.61 in 16D; 112.40 in PCA-5D) and Silhouette Score (0.2986 in 16D; 0.3945 in PCA-5D) while recovering 91.22% of true cultivar structure (ARI = 0.9122), and Full-Covariance GMM reaches ARI = 0.9637 with calibrated posterior membership probabilities.
2. **Stakeholder Interpretability via Centroid Profiles:** Unlike black-box manifold embeddings or fragmented DBSCAN sub-clusters, K-Means centroids and PCA loadings translate directly into physical laboratory thresholds (mg/L Proline, % Alcohol, g/L Flavanoids) that enologists and quality managers can audit on the production floor.
3. **Zero Unassigned Inventory in Core Tiering:** Whereas DBSCAN leaves 8.4% to 47.2% of batches unclassified as noise, K-Means and GMM assign 100% of production batches to actionable commercial tiers while GMM posterior entropy flags borderline batches.

Figure 3: 2D PCA Subspace Projection (PC1: 37.02% vs. PC2: 20.76% Variance) & Cluster Centroids



6. Key Findings and Actionable Stakeholder Insights

Cross-referencing the Champion K-Means (k = 3) centroids with PCA axes reveals **three highly distinct biochemical product archetypes** across the 178 batches (Table 3):

Key Biochemical Attribute	Cluster A: Reserve High-Body (N = 61 | 34.27% | Sil = 0.359)	Cluster B: Chromatic Structured (N = 49 | 27.53% | Sil = 0.355)	Cluster C: Delicate Aromatic (N = 68 | 38.20% | Sil = 0.204)
Alcohol (% vol)	13.68 (Z = +0.84)	13.15 (Z = +0.19)	12.28 (Z = -0.89)
Proline Amino Acid (mg/L)	1101.30 (Z = +1.13)	627.55 (Z = -0.38)	514.97 (Z = -0.74)
Flavanoids (g/L)	3.01 (Z = +0.98)	0.80 (Z = -1.24)	2.04 (Z = +0.01)
Total Phenols (g/L)	2.85 (Z = +0.90)	1.68 (Z = -0.99)	2.24 (Z = -0.09)
Color Intensity (abs)	5.47 (Z = +0.19)	7.27 (Z = +0.99)	3.04 (Z = -0.89)
Malic Acid (g/L)	1.97 (Z = -0.33)	3.33 (Z = +0.91)	1.93 (Z = -0.36)
Hue Ratio (420/520nm)	1.07 (Z = +0.49)	0.69 (Z = -1.20)	1.05 (Z = +0.42)
OD280/OD315 Protein Index	3.17 (Z = +0.78)	1.69 (Z = -1.30)	2.78 (Z = +0.23)
Engineered: Flavonoid Purity Ratio	1.05 (Z = +0.75)	0.49 (Z = -1.26)	0.91 (Z = +0.24)
Engineered: Chromatic Density Index	5.20 (Z = -0.19)	11.20 (Z = +1.23)	2.98 (Z = -0.72)

Table 3: Quantitative Centroid Profiles (Raw Units & Standardized Z-Scores) Across the Three Discovered Segments.

- **Insight 1 — Cluster A (Reserve High-Body & Polyphenol-Rich Tier, 34.3% of batches):** Anchored at PC1 = +2.50, PC2 = +0.93, Cluster A exhibits exceptional grape maturity with **Proline = 1,101.3 mg/L (Z = +1.13)**, **Alcohol = 13.68% (Z = +0.84)**, and **Flavanoids = 3.01 g/L (Z = +0.98)**. *Business Action:* Designate for premium cellar aging and flagship reserve pricing.
- **Insight 2 — Cluster B (Chromatic High-Acidity & Tannin-Depleted Tier, 27.5% of batches):** Located at PC1 = -2.99, PC2 = +1.25, Cluster B displays intense optical pigmentation (**Color Intensity = 7.27 abs, Z = +0.99; Chromatic Density = 11.20, Z = +1.23**) and sharp malic acidity (**3.33 g/L, Z = +0.91**), but severe depletion of Flavanoids (**0.80 g/L, Z = -1.24**) and OD280/OD315 (**1.69, Z = -1.30**). *Business Action:* Route to structured blending programs or oak-tannin supplementation to balance astringency.
- **Insight 3 — Cluster C (Delicate Low-Alcohol & Balanced Everyday Tier, 38.2% of batches):** Centered at PC1 = -0.09, PC2 = -1.73, Cluster C is defined by light alcohol (**12.28%, Z = -0.89**), low Color Intensity (**3.04 abs, Z = -0.89**), and bright Hue (**1.05, Z = +0.42**). *Business Action:* Position for immediate early-release retail distribution requiring zero cellar holding cost.

7. Model Limitations and Actionable Plan for Next Steps

Critical Evaluation of Methodological & Data Limitations:

1. **Spherical Voronoi Bias in K-Means:** Standard K-Means assumes isotropic, spherical clusters of equal variance in Euclidean space. Cluster C exhibits a wider dispersion (mean Silhouette = 0.2044 vs. 0.3587 in Cluster A), causing 6 boundary batches to sit near the equidistant Voronoi hyperplane between Clusters A and C.
2. **Euclidean Distance Concentration in 16D Space:** As demonstrated by our DBSCAN experiments, density estimation in 16 dimensions suffers from the Curse of Dimensionality (flagging 47.2% noise in 16D unless preceded by PCA reduction).
3. **Cross-Sectional Sample Size & Vintage Scope:** With N = 178 batches from a single regional cohort, the dataset lacks multi-vintage climatic drift, fermentation yeast telemetry, and consumer hedonic scorecards.
4. **Linearity Assumption of PCA:** Standard PCA only captures linear orthogonal correlations; curved biochemical saturation trajectories can project imperfectly onto flat hyperplanes.

Actionable Roadmap for Next Steps & Future Work:

- **Step 1 — Deploy Soft-Margin GMM Posterior Gating in Production:** Replace hard K-Means thresholding with Full-Covariance GMM posterior probabilities; batches with maximum cluster probability < 0.75 will trigger automated secondary lab verification.
- **Step 2 — Integrate Non-Linear Manifold Learning (UMAP + HDBSCAN):** Implement Uniform Manifold Approximation and Projection (UMAP) coupled with Hierarchical DBSCAN (HDBSCAN) to eliminate manual epsilon tuning and capture variable-density sub-clusters.
- **Step 3 — Expand Feature Space with Spectroscopy & Sensory Panel Targets:** Incorporate Fourier-Transform Infrared (FTIR) spectra, volatile aroma gas-chromatography (GC-MS), and multi-vintage harvest timestamps to monitor Population Stability Index (PSI) drift.
- **Step 4 — Operationalize Cluster-Specific Supervised Models:** Productionize the cluster-local Ridge/Gradient Boosting ensemble (which reduced prediction RMSE by 9.21%-10.48% in Section 4.3) inside an automated MLflow quality-scoring pipeline.